

regole

Regole di derivazione	
$y = f(x) \pm g(x)$	$y' = f'(x) \pm g'(x)$
$y = f(x) \cdot g(x)$	$y' = f'(x) \cdot g(x) + f(x) \cdot g'(x)$
$y = k \cdot f(x) \quad k \in \mathbb{R}$	$y' = k \cdot f'(x)$
$y = \frac{f(x)}{g(x)}$	$y' = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{[g(x)]^2}$
$y = f(g(x))$	$y' = f'(g(x)) \cdot g'(x)$
Funzione	Funzione derivata
$y = k \quad k \in \mathbb{R}$	$y' = 0$
$y = x^a \quad (CP: y = \sqrt{x})$	$y' = ax^{a-1} \quad (CP: y' = \frac{1}{2\sqrt{x}})$
$y = a^x \quad (CP: y = e^x)$	$y' = a^x \ln a \quad (CP: y' = e^x)$
$y = \log_a x \quad (CP: y = \ln x)$	$y' = \frac{1}{x \ln a} \quad (CP: y' = \frac{1}{x})$
$y = \sin x$	$y' = \cos x$
$y = \cos x$	$y' = -\sin x$
$y = \operatorname{tg} x$	$y' = \frac{1}{\cos^2 x} = 1 + \operatorname{tg}^2 x$
$y = \arcsin x$	$y' = \frac{1}{\sqrt{1-x^2}}$
$y = \arccos x$	$y' = -\frac{1}{\sqrt{1-x^2}}$
$y = \operatorname{arctg} x$	$y' = \frac{1}{1+x^2}$

$\lim_{h \rightarrow 0} \frac{f(x_0+h) - f(x_0)}{h} = f'(x_0)$
 DEFINIZIONE

1. Calcola le derivate delle seguenti funzioni, applicando le regole di derivazione e semplificando le espressioni ottenute:

$y = \arcsin(2x-3)$
 $y = \sqrt{\cos x}$
 $y = \log_5(x^4 - 3x^3)$
 $y = \frac{1}{(7x-3)^4}$
 $y = 10^{(2x-1)^2}$
 $y = \sin \ln\left(\frac{x}{2}\right)$

$y = 10 - 2x + x^3 - 2x^4$
 $y = \sqrt{2x^2 - 1}$
 $y = \operatorname{tg}\left(\frac{x}{2}\right)$
 $y = x + 4\sqrt{x} - 5 \operatorname{arctg} x$
 $y = \frac{6x^2 + 7x - 3}{2x+3}$
 $y = \sin^2(-2x)$
 $y = 4^x + \frac{\arcsin x}{2}$
 $y = \frac{5}{x} + \frac{1}{3x^2}$
 $y = (2 - 5x^2)^{20}$
 $y = (2x^2 - 1) \ln x$
 $y = x^2(4 - x^2)^2$
 $y = x \cdot e^x \cdot \operatorname{tg} x$
 $y = \frac{1-3x}{x^2-3}$
 $y = \frac{1+x^2}{\sin x}$
 $y = \operatorname{arctg} \sqrt{2x-1}$

esempi:

$$y = x^2$$

$$y' = 2x$$

$$y = 5x^2$$

$$y' = 10x$$

$$y = 10 - 2x + x^3 - 2x^4$$

$$y' = 0 - 2 + 3x^2 - 8x^3$$

$$y = x + 4\sqrt{x} - 5 \operatorname{arctg}(x)$$

$$y' = 1 + \frac{4}{2\sqrt{x}} - 5 \cdot \frac{1}{1+x^2} =$$

$$= 1 + \frac{2}{\sqrt{x}} - \frac{5}{1+x^2} = \frac{(1+x^2)\sqrt{x} + 2(1+x^2) - 5\sqrt{x}}{(1+x^2)\sqrt{x}} = \dots$$

$$y = \frac{5}{x} + \frac{1}{3x^4} = 5 \cdot \frac{1}{x} + \frac{1}{3} \cdot \frac{1}{x^4} = 5x^{-1} + \frac{1}{3}x^{-4}$$

$$y' = -5x^{-2} - \frac{4}{3}x^{-5}$$

$$y = \arcsin(g(x)) \quad f(x)$$

$$y' = 2 \cdot \frac{2}{\sqrt{1-(2x-3)^2}} = \frac{4}{\sqrt{1-4x^2+9-12x}} =$$

$$= \frac{4}{\sqrt{4x^2-12x+10}} = \dots$$

$$\sqrt{x} \quad 2 \text{ modi}$$

$$1) \frac{1}{2\sqrt{x}}$$

$$2) x^{1/2} \rightarrow \frac{1}{2} x^{1/2-1}$$

$$\arcsin(x) \rightsquigarrow \frac{1}{\sqrt{1-x^2}}$$

$$\arcsin(2x) \rightsquigarrow \frac{2}{\sqrt{1-4x^2}}$$

- $y = \sqrt{2x^2 - 1} = \left((2x^2 - 1)^{1/2} \right)^{g(x)} f(x)$

$$y' = 2x \cdot \frac{1}{2} (2x^2 - 1)^{1/2 - 1} = 2x \cdot (2x^2 - 1)^{-1/2} = \frac{2x}{\sqrt{2x^2 - 1}}$$

- $y = \left(\lg\left(\frac{x^2}{2}\right) \right)^{g(x)} f(x) = \lg\left(\frac{1}{2} \cdot x^2\right)$

$$y' = x \cdot \frac{x}{\cos^2\left(\frac{x^2}{2}\right)} = \frac{x^2}{\cos^2\left(\frac{x^2}{2}\right)}$$

$$\lg(x) = \frac{1}{\cos^2(x)}$$

$$\lg(2x) = \frac{2}{\cos^2(2x)}$$

- $y = \frac{6x^2 + 7x - 3}{2x + 3}$

$$y' = \frac{(12x + 7)(2x + 3) - (6x^2 + 7x - 3) \cdot 2}{(2x + 3)^2} = \frac{24x^2 + 36x + 14x + 21 - 12x^2 - 14x + 6}{(2x + 3)^2} =$$

$$= \frac{12x^2 + 36x + 27}{(2x + 3)^2} = \dots$$

- $y = \sin^2(\underbrace{-2x}_{g(x)}) = \sin(-2x) \cdot \sin(-2x)$

derivata di $\sin(-2x)$ è $-2 \cdot \cos(-2x)$
quindi risolveremo come prodotto di derivate

$$y' = -2 \cos(-2x) \sin(-2x) + -2 \cos(-2x) \sin(-2x) =$$

$$= 2(-2 \cos(-2x) \sin(-2x)) = -4 \cos(-2x) \sin(-2x)$$